Long-Term Support (LTS) Dependency Analysis & Migration Plan

Project: SV-UI (UPYOG Platform)
Document Type: LTS Upgrade Implementation

Created By: NIUA(NUDM)
Date: 16th April 2026

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1. Summary

This document describes the successful upgrade of the SV-UI frontend to make it **LTS compliant, secure, and compatible with modern React (19)**.

The upgrade includes migration of outdated dependencies, removal of deprecated APIs, and modernization of architecture.

2. Upgrade Summary

Area	Before	After
Yarn	1.22.19	1.22.22
Node.js	14.x	22.x
React	17.x	19.x
Routing	react-router-dom v5	v6
React Dom	17.x	19.0.0
Data Fetching	react-query v3	@tanstack/react-query v5
HTTP Client	Axios 0.21.x	Axios 1.x
Build Tool	CRA	VITE
Babel	babel-preset-react 6.x	@babel/preset-react 7.x
i18next	i18next 19.x	23.x
Forms	React Hook Form 6.x	7.x
Axios	0.21.x	1.x
React-i18next	11.x	14.x
Babel Preset React	babel-preset-react 6.x	@babel/preset-react 7.x
React Tooltip	V4	V5
Date Fns	2.30.0	3.6.0
Rechart	2.0.9	2.10.0

3. This library is divided into three parts:

- High Priority Libraries (Critical)
- Medium Priority Libraries
- Low Priority Libraries

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3.1 Hight Priority Libraries (Critical)

- **REACT** ROUTING DOM (V5 → V6)

Item	Current	Target	Impact
react-router-dom	**5.3.0**	**6.22.0+**	**Breaking API changes**

Required Code Changes

- switch → Routes
- Route component={} → Route element ={}
- useHistory() → useNavigate()
- history.push() → navigate()
- Redirect → Navigate

- **REACT** Query → TanStack Query

Current	Target
react-query 3.x	**@tanstack/react-query 5.x**

Key Changes

- from "react-query" → from "@tanstack/react-query"
- useQuery('key', fn) → useQuery({queryKey, queryFn})
- useMutation (fn) → useMutation({mutationFn})

- **REACT** CRA → VITE

Status	Recommendation
Deprecated	**Migrate to Vite**

CRA no longer receives security updates and is not suitable for LTS systems.

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- ## **REACT** Library

Current	Target
React 17	**React 19**

Key Changes

- useEffect must not return null or a promise
- key prop must not be passed via spread
- In React 19 passing key inside a spread object is a error

- ## **Node** Version

Current	Target
Node 14	**Node 22**

Key Changes

- Upgrade Node.js to **v22**
- Upgrade Yarn to the **latest version**
- Upgrade react-scripts from **3.6.1 → 5.0.1**
  (Note: This can be removed after complete migration to Vite)
- Replace Node version in all package.json files:
  "node": ">=14" → "node": ">=22"

- ## **Yarn** Version

Current	Target
Yarn **1.22.19**	1.22.22

- ## **Babel** Preset React

Current	Target
babel-preset-react 6.x	@babel/preset-react 7.x

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- **Axios**

Current	Target
0.21.x	1.x

- **REACT** ROUTING DOM

Current	Target
react-dom 17.x	react-dom 19.x

- **CREATE CUSTOM HOOK** For Navigation

  During the migration, a custom navigation hook was introduced to abstract routing logic and ensure smooth transition across the application.

  Instead of directly using useNavigate in multiple components, a centralized custom hook was created

  Custom Hook: useCustomNavigate

Path: web/micro-ui-internals/packages/libraries/src/hooks/useCustomNavigate.js

```
import { useNavigate } from "react-router-dom";

const useCustomNavigate = () => {
  const navigate = useNavigate();

  return (to, options = {}) => {
    try {
      navigate(to, options);
    } catch (error) {
      console.error("Navigation error:", error);
      // Fallback for navigation failures
      if (typeof to === "string") {
        window.location.href = to;
```

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```
    }
   }
  };
};

export default useCustomNavigate;
```

Path: web/micro-ui-internals/packages/libraries/src/hooks/index.js

```
// Register this hook in index.js
import useCustomNavigate from "./useCustomNavigate";
const Hooks = {
useCustomNavigate
}
```

How to use

```
const navigate = Digit.Hooks.useCustomNavigate();
```

1.1 Medium Priority Libraries

- **I18next**

Current	Target
i18next 19.x	i18next 23.x

- **React-i18next**

Current	Target
react-i18next 11.x	react-i18next 14.x

- **React Form Hook**

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Current	Target
react-hook-form 6.x	react-hook-form 7.x

1.2 Low Priority Libraries

- **Date**

Current	Target
date-fns 2.30.0.x	date-fns 3.6.x

- **React Tooltip**

Current	Target
V4	V5

- **Recharts**

Current	Target
recharts 2.x	2.0.9